eco-spec

Architectural Specification & Scientific Domain Integration Engine

Author: Independent RSE

Target: Max Planck Institute for Animal Behavior / Uni Konstanz

Status: Active Architecture Design

1. Executive Summary & Problem Statement

Interdisciplinary software development in behavioral ecology and psychophysiology often suffers from a severe semantic gap between domain experts (field biologists, ethologists, researchers) and software engineering teams. Requirements are traditionally gathered via unstructured text documents, verbal interviews, or static spreadsheets, leading to misaligned data models, costly refactoring cycles, and delayed project timelines.

> **Core Objective:** ***eco-spec*** is an open-source, highly decoupled Angular application designed to streamline requirements elicitation. It empowers scientists to visually configure survey specifications, processes them via a local Web Worker database pipeline, and instantly exports executable specifications (***JSON/YAML***) alongside human-readable documentation (***Markdown***).

2. Technical Architecture & Tech Stack

The system is built on a modern, reactive web architecture specifically optimized for offline-first field usability and instant client-side processing without heavy backend infrastructure dependency.

Layer	Technology / Paradigm	Rationale & Scientific Benefit
Frontend Engine	Angular (Signals, Resource API)	Guarantees fine-grained reactivity, eliminating unnecessary change detection cycles and managing complex form state smoothly.
Computation	Web Workers	Offloads heavy JSON schema parsing, schema validation, and export generation to background threads, maintaining a stutter-free UI.
Storage / State	IndexedDB / Local DB	Ensures local persistence of complex survey states, allowing researchers to draft, save, and iterate offline during fieldwork.
Interoperability	Dynamic JSON-driven Schemas	Decouples the UI rendering engine from scientific domain logic, enabling infinite adaptability across biological use cases.

3. Workflow & Data Pipeline

The interaction pipeline is engineered to minimize friction for academic personnel while establishing a rigorous contract for software engineers:

- **Step 1: Dynamic Questionnaire Generation:** Researchers load an adaptive web interface where form fields (telemetry frequency, spatial resolution, behavioral categories, sensor sampling rates) are rendered dynamically from modular JSON schemas.
- **Step 2: Asynchronous Validation:** Form input and constraints are validated in real-time inside a dedicated Web Worker thread.
- **Step 3: Dual-Artifact Export:** Upon completion, the system generates:
  - *JSON / YAML Schema:* Machine-readable specifications ready for backend ingestion or automated code scaffolding.
  - *Markdown Report:* A structured summary formatted for direct insertion into Jira tickets, GitHub issues, or collaboration wikis.

4. Mathematical Modeling of Field Constraints

Data acquisition in telemetry engines requires strict parameter bounds to prevent sensor battery depletion and memory overflows in remote devices. *eco-spec* models schema constraints dynamically:

$$S_{total} = f_{sample} \times t_{duration} \times (size_{packet} + overhead_{meta}) \leq Limit_{storage}$$

Where f_{sample} is the sampling frequency defined by the biologist in the questionnaire, $t_{duration}$ is the deployment window, and S_{total} represents expected data volume. The engine validates these parameters against hardware limitations before generating the final developer specification.

5. Strategic Impact on Scientific Collaboration

By deploying *eco-spec* prior to technical alignment meetings with research groups (such as those at the Max Planck Institute for Animal Behavior), development cycles are compressed from weeks of ambiguous discussions into concrete, testable prototypes. This demonstrates proactive systems architecture design and bridges academic research with high-performance software engineering.